21/9/2026

Practical-8

Aim: To discover Live Hosts Using Nmap Scans (ARP, ICMP, TCP/UDP) on the TryHackMe Platform.

Intro - NMAP is a free, open source network-scanning tool created by Gordon Lyon.
It can determine whether a host is alive using four protocols.

- ARP - ICMP - TCP - UDP

Setup:

1. Open the room: https://tryhackme.com/room/nmap01 & log in to your TryHackMe account.
2. Use the embedded network simulator to answer the conceptual questions.
3. Click start to deploy the target network - a small lab with one Windows host & one Ubuntu host.
4. Click ~~initialise~~ AttackBox has booted, open terminal inside it.
5. note the CONNECTION. IP/target

ARP discovery:

$ sudo nmap -PR -sn 10.200.6.0/24

O/P

Starting Nmap 7.80 (https://nmap.org)
Nmap scan report for 10.200.6.1
Host is up

Nmap done: 256 IP addresses (1 host up) scanned in 10.51 seconds

ICMP Host Discovery

(i) ICMP Echo (-PE)

$ sudo nmap -PE -sn 10.200.6.0/24
Nmap scan report for 10.200.6.1
Host is up (0.069s latency).
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addresses (3 host up) scanned in 3.38 seconds

(ii) ICMP Timestamp (-PP)

$ nmap -PP -sn 10.200.60.0/24
Nmap scan report for 10.200.6.50
Host is up (0.069s latency)
Nmap done: 256 IP addresses (2 host up) scanned in 8.44 seconds

(iii) ICMP Address Mask (-PM)

$ nmap -PM -sn 10.200.6.0/24
Nmap done: 256 IP addresses (0 hosts up) scanned in 52.14 seconds

TCP & UDP Ping Discovery

(i) TCP SYN Ping (-PS)

$ nmap -PS -sn 10.200.6.0/24
Nmap scan report for 10.200-6.50

Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.068s latency)
Nmap done: 256 IP addresses (2 hosts up)
Scanned in 33.88 seconds

(ii) TCP ACK Ping (-PA)

$ sudo nmap -PA -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency)
Nmap done: 256 IP addresses (2 hosts up) scanned in 36.64 seconds

(iii) UDP Ping (-PU)

$ sudo nmap -PU -sn 10.200.6.0/24
Nmap scan report for 10.200.6.50
Host is up (0.070s latency)
Nmap scan report for 10.200.6.250
Host is up (0.069s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 35.97 seconds

Nmap's Default Host-Discovery Behaviour

- Privileged user, local Ethernet target → ARP requests
- Privileged user, target outside the local network, ICMP Echo, TCP ACK to port 80

- Unprivileged user, target outside the local network → TCP 3-way handshake to ports 80 & 443

Masscan - transmits packets far more aggressively, trading some accuracy for raw speed on large ranges.

$ masscan 10.200.6.0/24 -p443
$ masscan 10.200.6.0/24 -p80,443
$ masscan 10.200.6.0/24 -p22-25

Reverse DNS Lookup
resolves an IP address back to a hostname.

Steps:

1) Log in TryHackMe, open "Nmap Live Host Discovery" room

2) Work through task 1-3 & answer the questions

3) Deploy the lab network in task 4 & start Attack Box, note the target subnet / CONNECTION_IP.

4) Run an ARP scan: sudo nmap -PR -sn <subnet>/24 & record how many hosts respond

5) Run an ICMP Echo scan: sudo nmap -PE -sn <subnet>/24 - compare

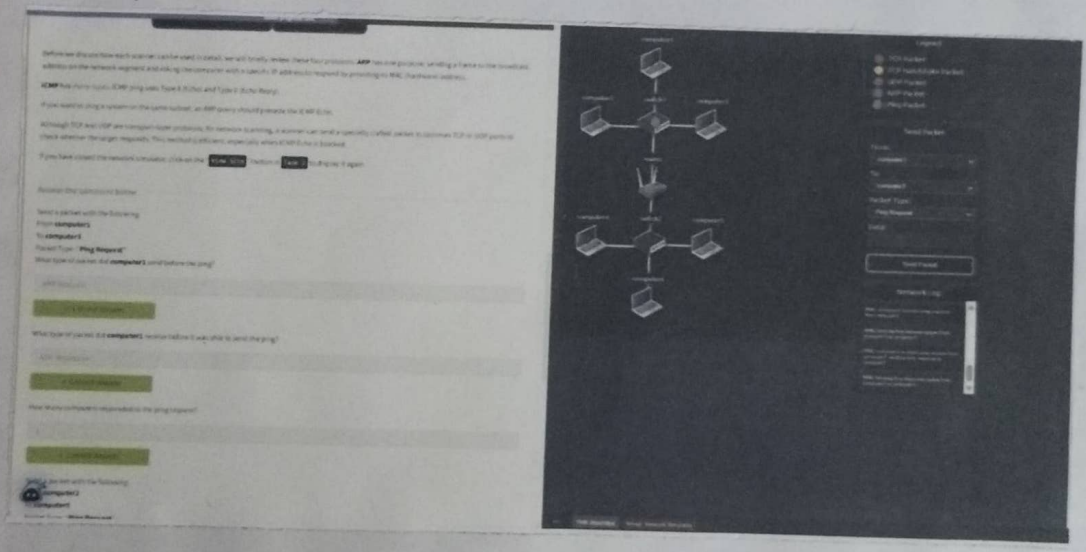
the result with ARP scan.

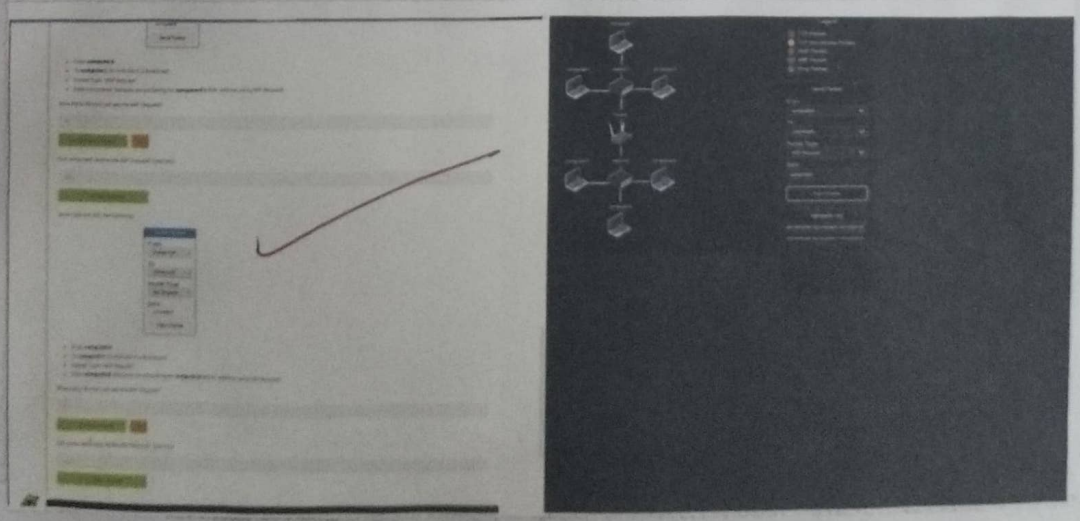
6) Run ICMP Timestamp (-PP) & ICMP Address Mask (-PM) scans.

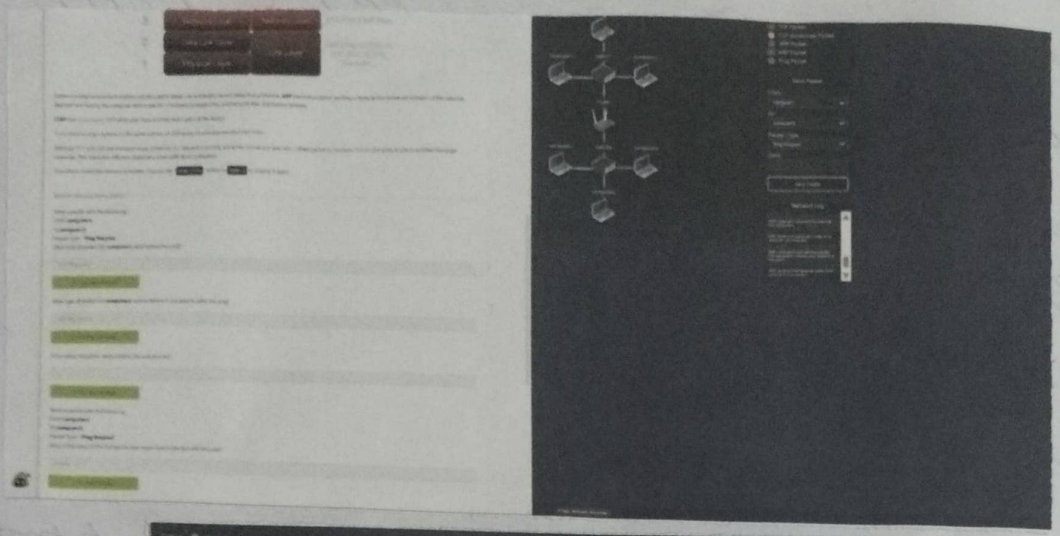
7) Run a TCP SYN ping (-PS), a TCP ACK ping (-PA, needs sudo) and a UDP ping (-PU), compare results & required privilege level for each.

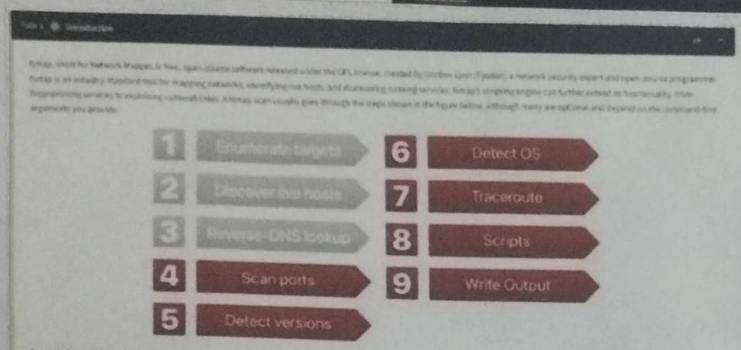
8) Try nmap -sL <subnet>/24 & nmap -R <subnet>/24

9) Complete room's questions & move to Nmap Basic Port Scans









1 Enumerate targets
2 Discover live hosts
3 Reverse-DNS lookup
4 Scan ports
5 Detect versions
6 Detect OS
7 Traceroute
8 Scripts
9 Write Output

Learning Objectives

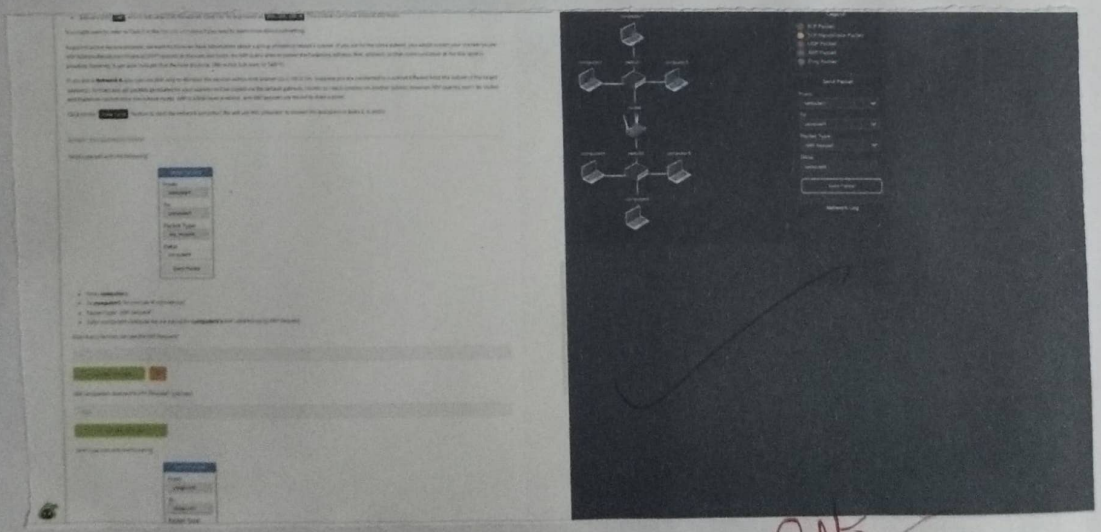
1. Which systems are up?
2. What services are running on these systems?

- Understand why host discovery is a necessary first step before port scanning, and why scanning offline systems wastes time and generates unnecessary noise.
- Learn how ARP scans work at the link layer, using broadcast requests to identify live hosts within the same subnet.
- Understand how ICMP echo, timestamp, and address mask requests can be used to discover live hosts across different network segments.
- Learn how TCP SYN and ACK ping scans use transport layer packets to detect online hosts, and understand the difference between privileged and unprivileged scan behaviour.
- Understand how UDP ping scans trigger ICMP port unreachable responses from closed ports to confirm a host is online.

Learning Prerequisites

In the next task, let's dig deeper and understand some important concepts before discovering live hosts.

Some of these questions will require a static site to answer the task questions, while others will require the attacker.



Result: Hence, Live Hosts were discovered successfully using NMAP scans on the TryHackMe Platform.